

- 7 (a) Define the term respiratory quotient (RQ).

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..... [2]

- (b) Fig. 7.1 summarises the respiration of an organic substance, **A**, in aerobic conditions.

- (i) Complete Fig. 7.1 in the spaces provided.



Fig. 7.1

[2]

- (ii) Calculate the RQ for substance **A**.

Write your answer to **two** decimal places.

RQ = [1]

- (iii) Use your calculated RQ value to suggest the group of compounds to which substance **A** belongs.

..... [1]

- (c) Dinitrophenol (DNP) is a compound used in the production of a number of products, including chemical dyes, insecticides and wood preservers. People who work in factories with DNP are at risk of exposure if they do not follow health and safety guidelines.

DNP is a molecule that can transport protons across the inner mitochondrial membrane, allowing protons to leak out of the intermembrane space.

- (i) Suggest **and** explain the effects of DNP on aerobic respiration in human cells.

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..... [4]

- (ii) Suggest **two** symptoms that may be experienced by people after exposure to DNP for several months.

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..... [2]

[Total: 12]